

Groups, Arrays, and Products

Turning repeated addition into a multiplication, reading an array as rows of equal size, writing a product back as repeated addition, and finding a missing factor

Three ways to show the same amount.

- **Equal groups added again and again:** $5 + 5 + 5$.
- **An array** — rows that all hold the same number of things.
- **A multiplication:** 3×5 means *3 groups of 5*.

They all give the same total. Show how you found each answer.

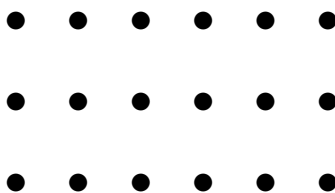
- 1.** Ana counts her stickers by fives:

$$5 + 5 + 5 + 5$$

Write this as a multiplication, then find the total.

Answer: _____

- 2.** Count the dots in the array. Write the multiplication that matches it, then give the total.



Answer: _____

3. Write $7 + 7 + 7$ as a multiplication, then find the total.

Answer: _____

4. This array has 4 rows with 7 stars in each row.



Write the multiplication for the rows, and then the multiplication you get by counting the columns instead. Give the total.

Answer: _____

5. Marco has 6 bags with 3 marbles in each bag. Write 6×3 as a repeated addition, then find how many marbles he has.

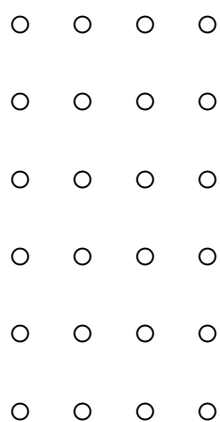
Answer: _____

6. Fill in the missing factor, then say how you know.

$$4 \times \underline{\quad} = 24$$

Answer: _____

7. A tray of muffins has 6 rows with 4 muffins in each row.



Write the multiplication and find how many muffins are on the tray.

Answer: _____

- 8.** Nia jumps along a number line, 6 at a time:

$$6 + 6 + 6 + 6 + 6$$

Write the multiplication that matches her jumps, and find where she lands.

Answer: _____

- 9.** Fill in the missing factor.

$$\underline{\hspace{1cm}} \times 5 = 35$$

Then finish this sentence: that many groups of 5 makes 35.

Answer: _____

- 10.** Ben was asked for 4×6 . He wrote $4 + 6 = 10$. Explain what Ben mixed up, write 4×6 as a repeated addition, and give the correct product.

Answer: _____

11. An array has 4 rows with 6 tiles in each row. Sam adds *one more row* of 6 tiles.

(a) How many rows are there now? (b) How many tiles are there altogether? Show the multiplication you used.

Answer: _____

12. Challenge. Kai has 32 tiles. He puts them in an array with 8 tiles in each row.

(a) How many rows does he make? (b) Write the repeated addition that matches that array. (c) Kai then rebuilds the array with only 4 tiles in each row. How many rows does he need now?

Answer: _____